

3 deg. of freedom $\rightarrow$ 6 dimensional phase space, Model a star in a galaxy with time dependent potential

$(R, \theta, z, \dot{R}, \dot{\theta}, \dot{z}) \rightarrow I_j(R, \theta, z, \dot{R}, \dot{\theta}, \dot{z})$, $j \in 1, \dots, 5$ see: Contopoulos 1957, 1958, 1960, 1963
Burbanck 1962,
Vandekerkhof 1962, 1963
Ollongren 1962

with $I_j = C_j$, $C_j = \text{constant}$

- C_j each representing a hyper surface, intersection become trajectories
- $I_j = C_j$ can be non-isolating (def. Vintner 1947; Lynden-Bell 1962; Ollongren 1962)
 - A nonisolating integral is such that the corresponding hypersurface consists of an infinity of sheets which fill the phase space densely.
 - $\Rightarrow I_j = C_j$ does not give any information for practical purposes, i.e. no conditions
 - $\rightarrow$ physics perspective, not significant, get ignored
 - $\rightarrow$ isolating integrals: integrals
 - $\rightarrow$ non-isolating integrals: —

present case $j=1, 2$ $I_1 = U_j(R, z) + \frac{1}{2} (\dot{R}^2 + R^2 \dot{\theta}^2 + \dot{z}^2)$ where do they come from though?
 $I_2 = R^2 \dot{\theta}$

I_4 & I_5 are generally non-isolating
a discussion around I_3 : isolating or non isolating

"But the assumption as has been often..."